Economics of Health and Health Care
Lecture 8: Imperfect information

Pedro Pita Barros
Faculdade de Economia
Universidade Nova de Lisboa

Other effects of uncertainty and asymmetric information

Without strategic use of information

- Information and increasing monopoly
- Small area variations

With strategic use of information

- Agency relationship
- Supplier-inducement demand

Information and increasing monopoly

No strategic use of information – uncertainty affects the flow of information

The lack of information over the prices and the quality of providers may imply that:

Entry of further providers - increase in competition - gives origin to higher prices

Theory: Satterthwaite 1979, Empirical evidence: Pauly e Satterthwaite 1981

Increasing monopoly

Lack of information - people trust on word-of-mouth, friends' advice and any information they collect

Health as a reputation good - differentiated product, information is disseminated by informal channels

Satterthwaite (1979): under monopolistic competition and reputation good, more firms means higher prices

Increasing monopoly

First implication: if the number of physicians in a given geographical area increases, then the information of consumers on each doctor decreases

Second implication: searching for a new doctor is more costly, consumers become less price sensitive.

Increasing monopoly

Consequence: an increase in the number of physicians may generate an increase in the price of medical servics.

Intuition: the smaller information that consumers have will give each physician a higher market power.

Empirical test: Pauly and Satterthwaite 1981

Increasing monopoly

The effect results from the lack of information and the breadth of that information

It explains why we may observe higher prices in larger cities than in smaller ones.

Of course: with informed consumers, we recover the usual economic effects.

Increasing Monopoly

Alternative assumption for same empirical observation: target income

More physicians, less income for each one, increase in services provided to keep income levels.

In this theory, consumers have a passive role, while in the increasing monopoly theory they have an active role.

We can use this distinction to test the theories.

Empirical test

Regression of the average price in a region on:

- Information variables - like density of doctors per geographic area, ease of transportation, social stability
- A measure of contacts; number of doctors per capita, as a supply measure; wages in industry (opportunity cost)
- Other demand side variables and control for other effects.

Increasing monopoly

The results without the information variables support the theory of target income - physicians per capita is positively correlated with average price

However, the information variables are statistically significant, which favours the interpretation of increasing monopoly vis-à-vis the theory of target income.

Increasing monopoly

Other explanations: opportunity cost of time - different mix

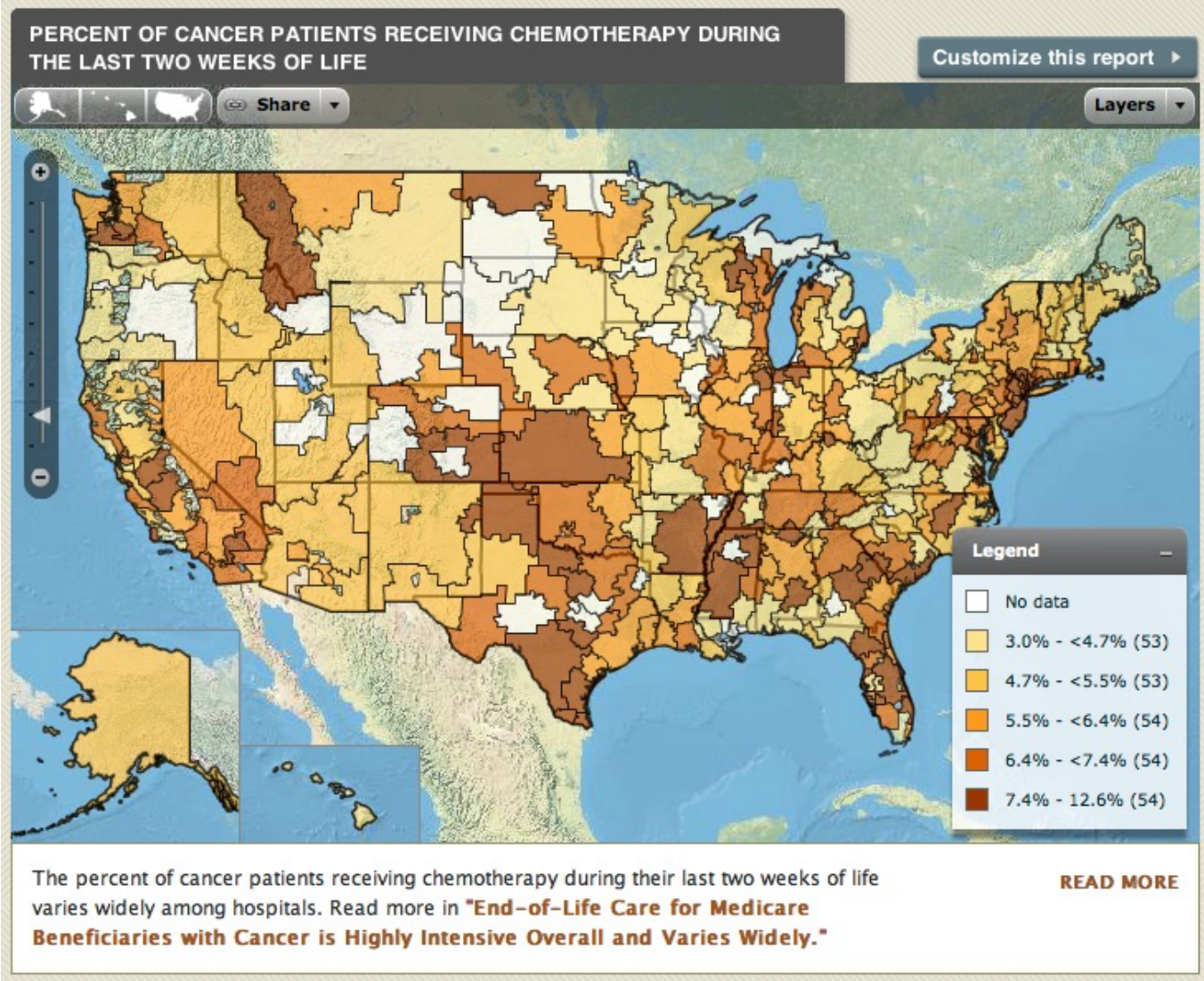
We do not have still a fully satisfactory answer:

the information problem is sufficiently important to revert the standard result that an increase in competition decreases prices?

How and why consumers obtain and use information?

Information: Small area variations

- This issue arises from the observation that considerable variation in clinical practice exists
- Doctors seem to differ in their decisions
- This variation is not conditional on characteristics of patients
- What motivates this variation?
- Should we be worried about?



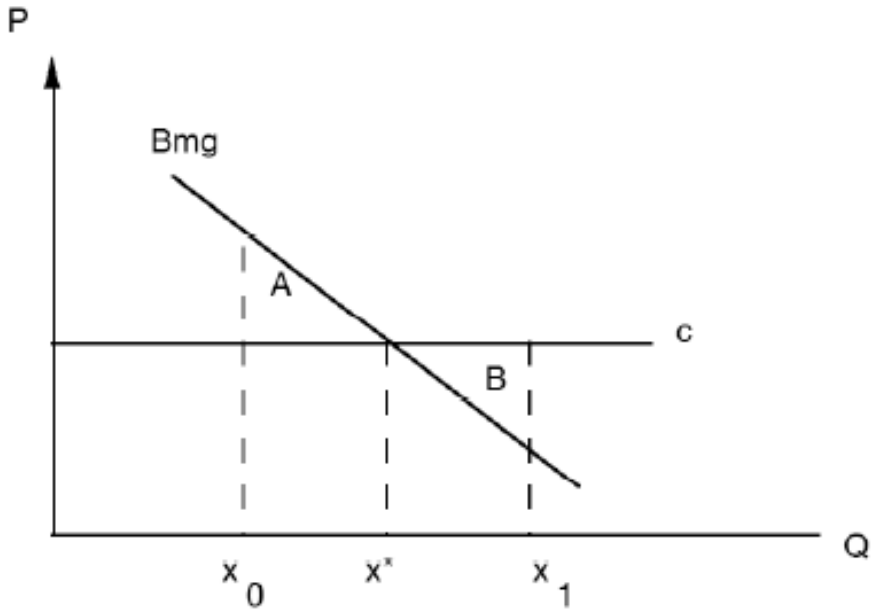
http://www.dartmouthatlas.org/

Practice variation: does it matter?

- Can then information be answer?
- Clinical guidelines and evidence-based medicine

- But should we worry?
- What is the cost of dispersion?

How to measure costs of practice variation?



- Cost of dispersion:

$$\Delta W = (p_1 - c)\frac{x^* - x_0}{2} + (c - p_2)\frac{x_1 - x^*}{2}$$

- With information on demand curve

$$(p_1 - c) \approx \frac{dp}{dx}dx = \frac{dp}{dx}(x_1 - x^*)$$

$$(c - p_2) \approx \frac{dp}{dx}(x^* - x_0)$$

$$\Delta W = \frac{1}{2}\frac{dp}{dx}\left[(x^* - x_0)^2 + (x_1 - x^*)^2\right]$$

$$\Delta W = \frac{1}{2}\frac{dp}{dx}\sum_i\left[(x^* - x_i)^2\right] \quad \Rightarrow \quad \Delta W = \frac{1}{2}\frac{dp}{dx}N\sigma_x^2$$

$$\Delta W = \frac{1}{2} p x^* N \frac{dp}{dx} \sigma_x^2 \frac{N}{x^{*2}} \frac{1}{\varepsilon} = \frac{p x^*}{2} \frac{N \sigma_x^2}{E(x)^2} \frac{1}{\varepsilon}$$

$$\Delta W = \frac{1}{2} \frac{(\text{total spending}) \times (\text{coeff of variation})}{\text{price elasticity}}$$

- Estimates of costs of small area variations
- Phelps and Parente (1990), Phelps and Mooney (1992): Bypass - 740 million USD

- More recent work: Finkelstein, Gentzklow, Williams (2016)
- What is the pattern of use of Medicare beneficiaries who move between regions with high and low average health care utilization
  - demand is “fixed” (same people), utilisation is the same
  - Use is determined by supply side, migrants will match the places of destination (not own characteristics)

results

- 40-50% comes from demand factors (health status, age, preferences)
- 50-60% comes supply factors (resources, practice styles, market structure)
- Type of services matter: emergency services variations are due to demand; imaging and diagnostic tests variations are due to supply.
- unobserved health differences and preferences matter
- No claims about efficiency of observed variation

Chandra and Staiger (2019) different view:

- Hospitals with greater expertise in a treatment use it more because of their comparative advantage => supply-side motive
- They found evidence of substantial variation in comparative advantage across hospitals: benefits from treatment being much higher in some hospitals than others
- Thus, treat more in some places than in others

- strict treatment guidelines are incorrect
- reducing treatment variation across hospitals may not be optimal
- Question: what is the origin of comparative advantage? Still to be answered

Implications

Agency relationship: What is it?

- Asymmetry of information leads to delegation of decisions
- Follow the principle that the best informed party should decide
- Ex: patient delegates on the doctor the selection of treatment
- If the interests of both coincide, that's fine

Agency relationship: What is it?

- What happens if they don't?
- Doctors may have different objectives and conflicting objectives relative to patients
- Think about a second visit fully paid by the patient
  - Doctors may want it (more income)
  - Patients may dislike it
  - Patients do not know if the visit is needed or not

Agency relationship

- There is an extensive literature in economics looking at delegation of decision under asymmetry of information
- We will look at the basic elements of the economic trade-offs involved in this delegation

Agency relationship

- Uncertainty: the health outcome of the patient is the result of combining the effort/skill of the doctor with pure random elements
- The asymmetry comes from the better knowledge of the doctor
- Thus, the patient cannot assess whether the doctor took the right decision or not
- The use of the privileged information, the effort the doctor devotes to the case cannot be observed

Agency relationship

- Something that cannot be observed cannot be used for payment
- Why is this relevant?
- Supose the doctor receives a fixed wage, independently of the final outcome.
- If doing more effort has higher costs to the doctor, what is the optimal decision for him?
  - Doing the least effort possible, of course.

Agency relationship

- Alternative: make the payment conditional on the final health status of the payment
- Now, a higher effort of the doctor has some reward, he will develop more effort
- At the same time, he faces more risk
- The higher incentive for effort goes hand in hand with higher risk

Agency relationship

- This will be a general characteristic: under asymmetric information, providing incentives implies usually imposing risk
- The trade-off is quite general. It exists even if doctors have some degree of altruism,
- It also applies to many other relations in health care.

Agency relationship

- Culyer (1989): a doctor is a perfect agent if he/she makes the same choice as the patient, would the patient have the same information
- The definition is close to the what we find on the ethical code of doctors (Hippocrates Oath)
- Which, actually, addresses the underlying economic tensions...

Hippocrates Oath

You do solemnly swear, each by whatever he or she holds most sacred.

That you will be loyal to the Profession of Medicine and just and generous to its members. That you will lead your lives and practice your art in uprightness and honor.

That into whatsoever house you shall enter, it shall be for the good of the sick to the utmost of your power, your holding yourselves far aloof from wrong, from corruption, from the tempting of others to vice.

That you will exercise your art solely for the cure of your patients, and will give no drug, perform no operation, for a criminal purpose, even if solicited, far less suggest it.

That whatsoever you shall see or hear of the lives of men or women which is not fitting to be spoken, you will keep inviolably secret.

These things do you swear. Let each bow the head in sign of acquiescence And now, if you will be true to this, your oath, may prosperity and good repute be ever yours; the opposite, if you shall prove yourselves forsworn.To this, your oath, may prosperity and good repute be ever yours.

Agency relationship

- The tension from payment to incentives has long been recognized
- The question is to best structure payments systems taking into account this tension

Supplier-induced demand

- Definition:
  - physicians may take advantage of asymmetric information to convince their patients to consume more medical care than would be in their self-interest
- This is a departure from the perfect agency story

Supplier-induced demand

- Several questions:
  - Is there evidence of supplier-induced demand?
  - Is there a natural limit to demand inducement?
  - Can economic design of relationships help on this matter?

The beginning

- Shain and Roemer (1959); Roemer (1961):
  - Found a strong correlation between hospital beds and its use - if a hospital has more beds it will try to fill them
  - The increase in use was detected after an increase in supply

Possible explanations

- Nominal rigidity of wages? (but then how is demand and supply balanced?)
- Target income theory - Rice (1983)
  - Doctors aim at a certain income, if for some reason it decreases, they will change activity levels - induce demand - to attain again the same income level

Potential problems

- How is determined the level of demand inducement?
- Why don’t doctors induce demand always the highest possible level?
- Evans (1974) - introduced ethical arguments, taken up by McGuire and Pauly (1991)
- Dranove (1988) - introduced reputation mechanisms - high inducement today is punished by lower demand tomorrow (but then patients need to be able to observe inducement ex-post)

McGuire-Pauly model

- They recognise that inducement has ethical costs for the doctor
- But it has financial advantages.
- So, they want to use the trade-off between financial gains and ethical costs to explain inducement
- And to see under whether a higher prices actually leads to inducement

The McGuire-Pauly Model

- Utility function of doctors: consumption, leisure and ethical desutility from demand inducement

$$U = U(P; L; I) = U_1(P) + U_2(L) + U_3(I)$$

- Income is determined by

$$P = mX(I)$$

- Time constraint:

$$L = 24 - tX(I)$$

- Set maximization problem and get first order condition:

$$\partial U/\partial I = U_1' m X'(I) + U_3' - tX'(I)U_2' = 0$$

- What is the impact of an increase in the price of medical care?
- If the utility is not too concave, inducement increases
- For target - income

$$\partial(mX(I))/\partial m = 0$$

Supplier-induced demand

- There are two weaknesses in the model,
  - The utility function is additive - interaction terms will make predictions of the model more confusing
  - The same effects can be obtained without the ethical considerations - the desutility of work is sufficient to generate the effects

Empirical evidence

- Need to distinguish movements along the demand curve from shifts of the demand curve
- Gruber and Owings (1996) - exogenous change in conditions, see reaction to it
- Exogenous change: decrease in fertility rate in the US

Empirical evidence

- Decision variable: do or not cesarean section in child delivery
- Gruber and Owings: a decrease of 13,5% in fertility lead to an increase in the use of cesarean section
- Price of cesarean: 2053 USD
- Price of natural childbirth: 1492 USD

- Adaptation of the McGuire-Pauly model

$$U=U(P, I)$$

- P- income, I - inducement $P=pn\ N + pc\ C$
- N- natural birth, C - cesarean section
- pn and pc - prices
- $I=B\ i$, B - total number of births,
- i - inducement per birth

$$C = a(i)\ B;\ N = (1-a(i))B$$

Empirical evidence

- The doctor chooses *i* to maximize utility
- Get the first order conditions and do comparative statics to obtain:
  - decrease in B increases inducement
  - decrease in B increases the share of cesarean section child delivery
- Leads to regression of CSEC (1/0) on demographic characteristics, previous cesarean case, characteristics of the hospital, number of professionals.

- Two variables to control for cesarean section:
  - Number of doctors (obst and ginecologists) per birth
  - Births per 100 persons

More doctors per birth, more Inducement is expected, Higher share of cesareans.

Lower fertility rate Higher inducement Higher share of Cesareans.

LOBKID	0,578			
	(0,126)			
LFERT		-0,888	-0,923	
		(0,202)	(0,201)	
LGEN		0,342		0,410
		(0,174)		(0,174)

Is this a US problem only?

- Evidence also from Europe, from national health systems
- Grytten, Carlsen and Sorensen (1995)
  - No evidence of demand inducement for visits
  - Evidence of inducement for tests
- Sorensen and Grytten (1999) - primary care in Norway - no evidence of inducement
- Van de Voorde et al (2001) - change in the insurance reimbursement in Belgium: Doctors are paid per case; consumers pay the full price and then receive back a fixed amount; This system has clear gains from inducement to doctors

To sum up, on inducement:

- Still need of further evidence, despite the apparently compelling argument
- Role of repeated relationship to control inducement?
- Information-based theories - how does the patient comply? Demand impedement (does not demand care as it suspects of inducement)